Report 3

TRAINDRAIN

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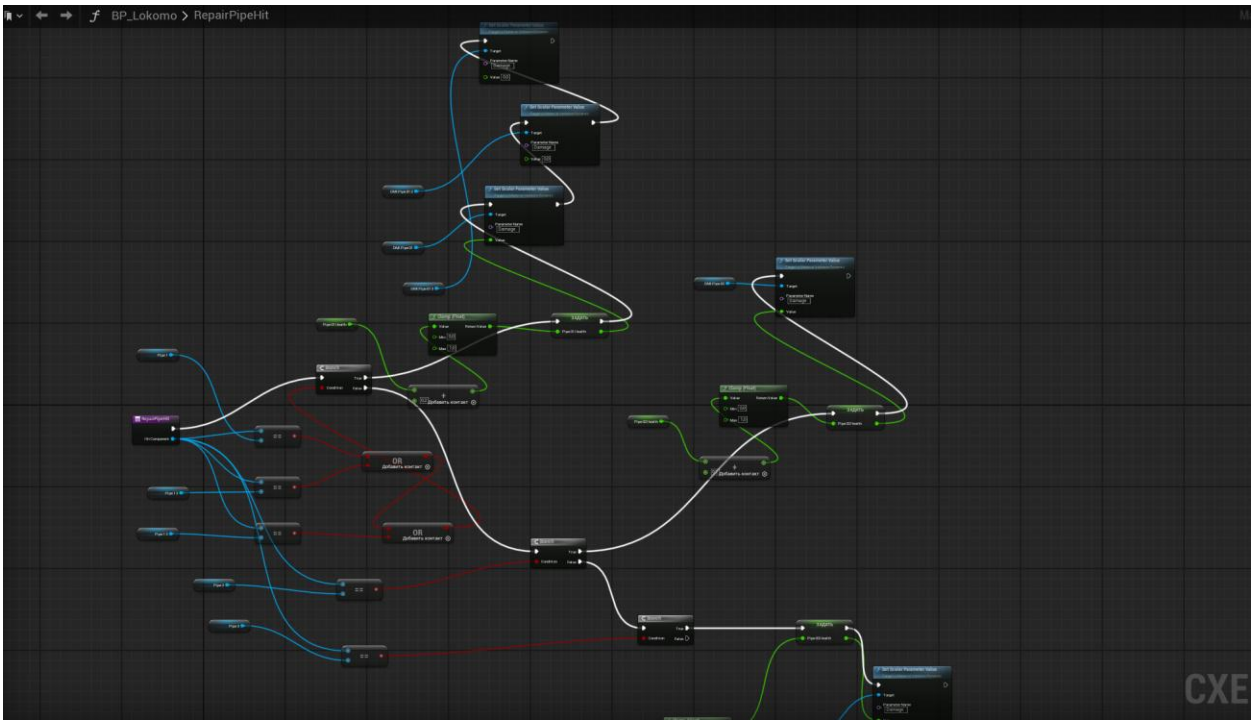

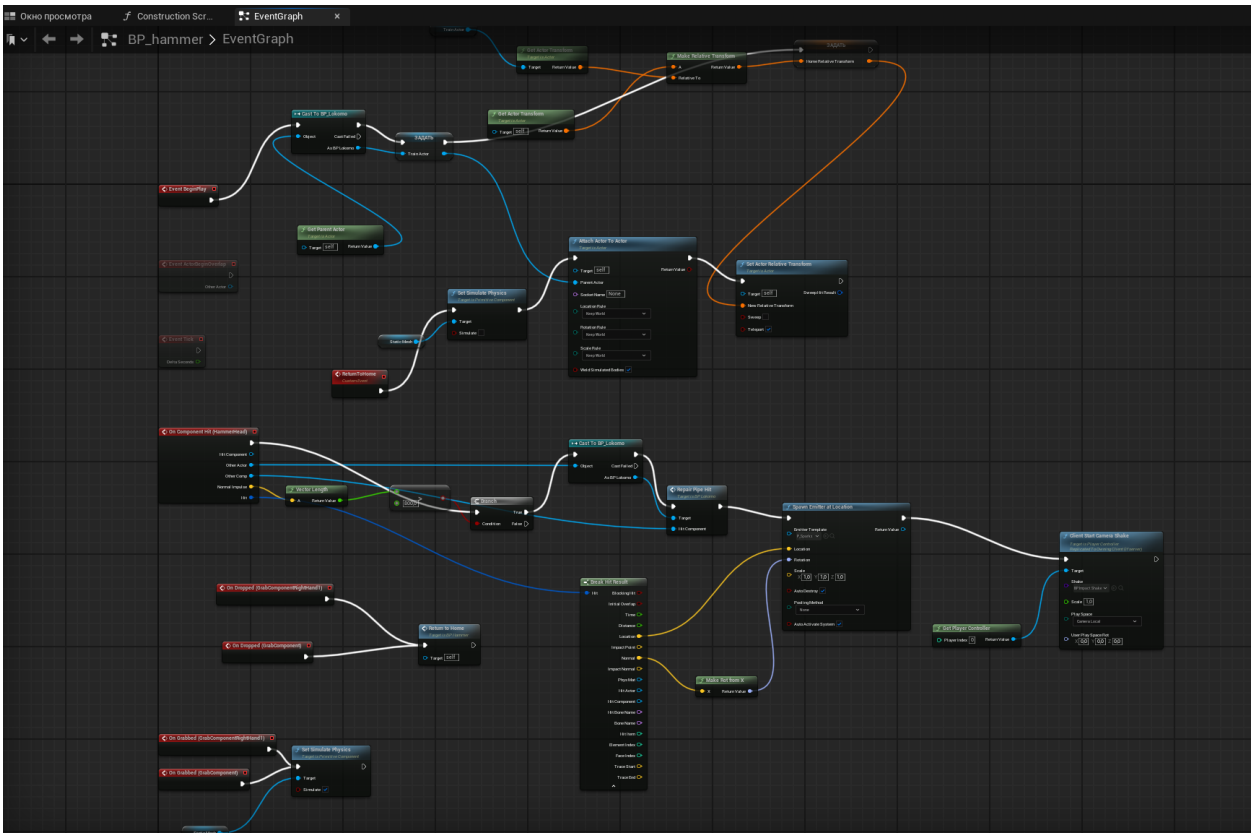

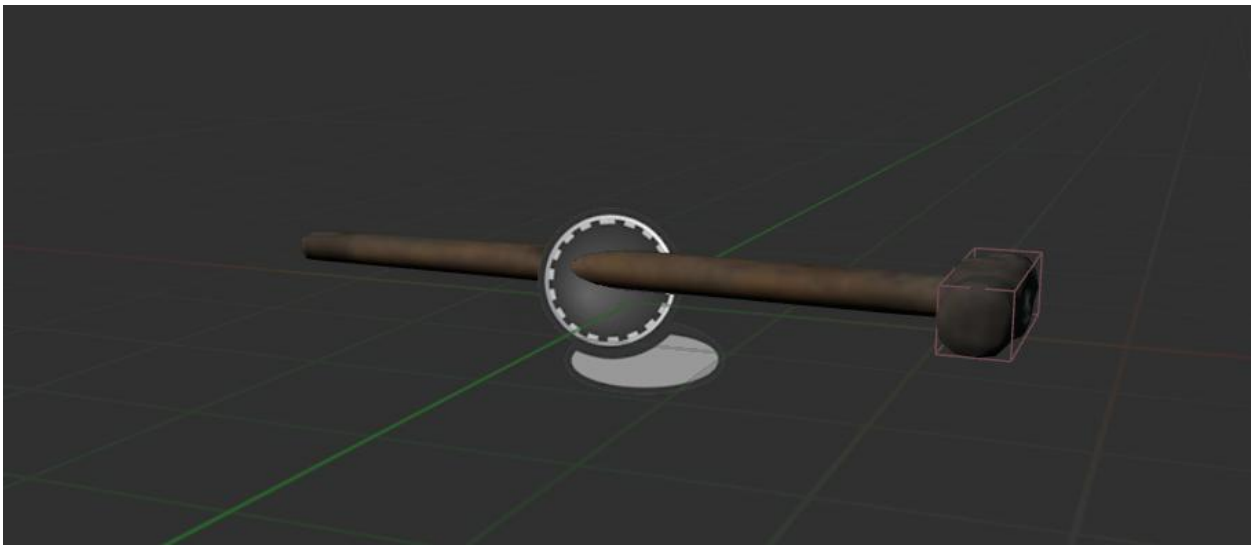

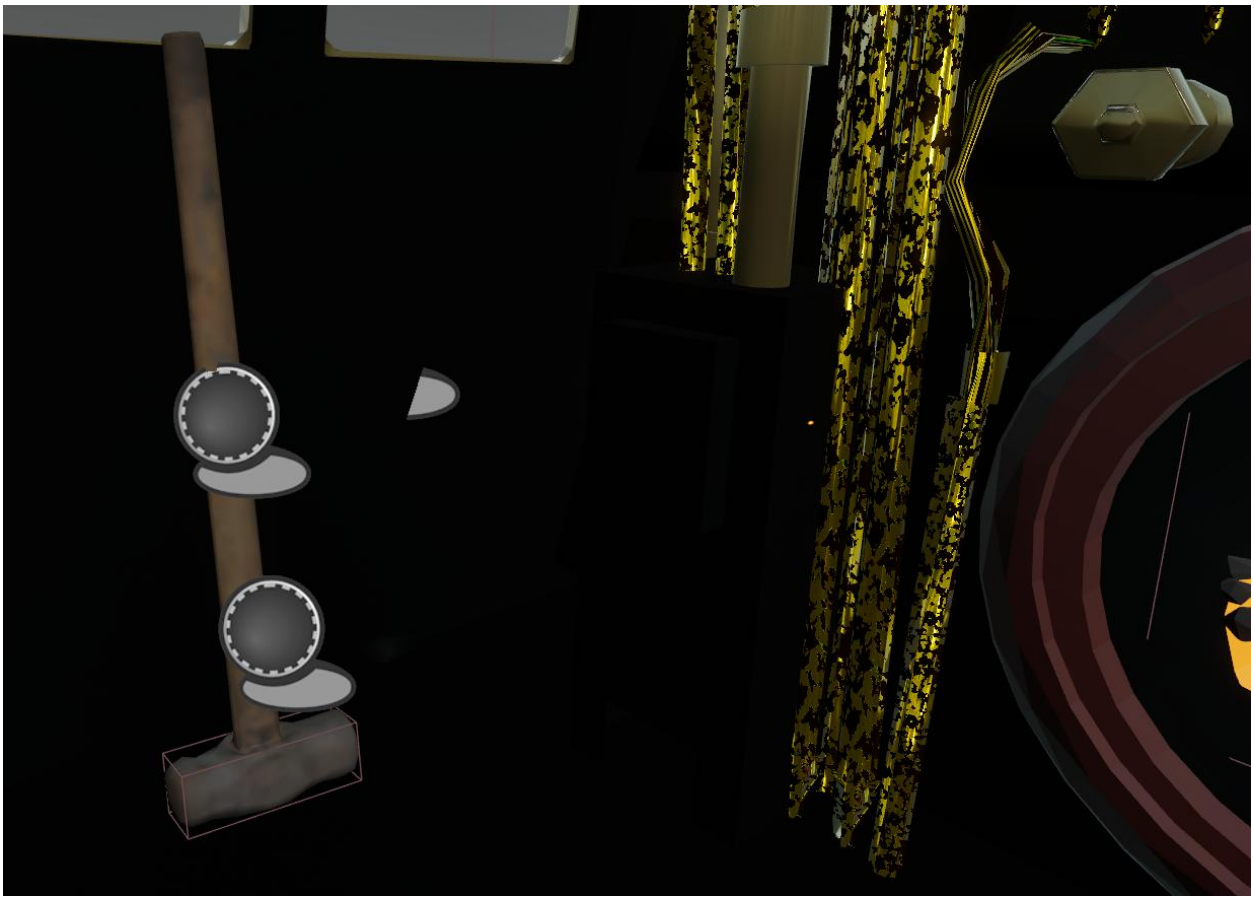

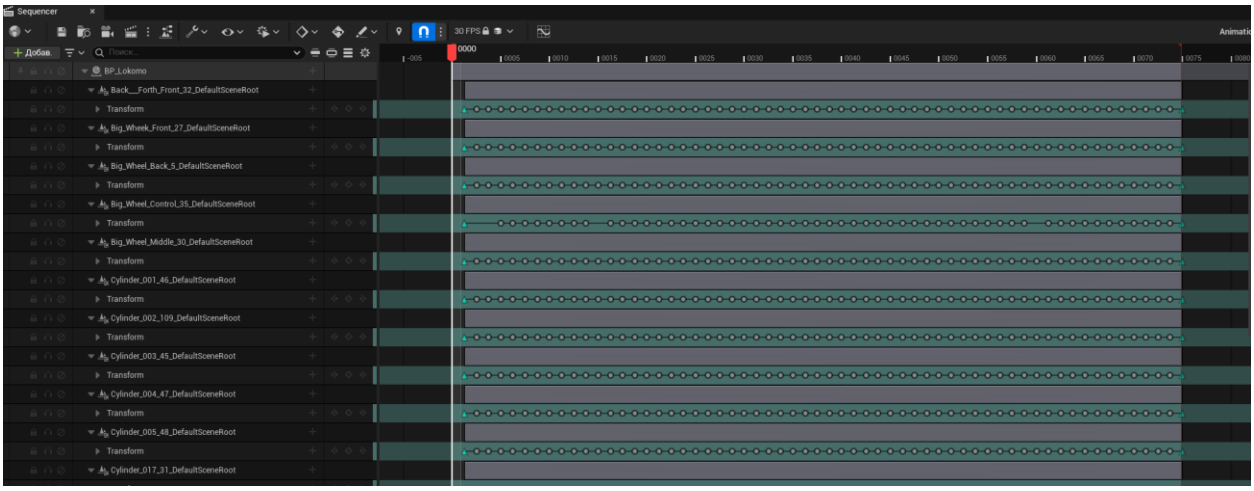

Current status

Game design

TrainDrain remains focused on driving the train as far as possible, but the internal train mechanics have been expanded with pressure/pipe integrity and overheating systems.

• A pipe pressure/integrity system has been implemented and made interactive: the player

must maintain the pipes in good condition.
• Coal efficiency is now tied to pipe condition: damaged pipes reduce how effective coal is for maintaining speed and overall acceleration.
• Maximum train speed now dynamically decreases as pipe condition worsens.
• Overheating is now a core risk factor: excessive coal input causes the engine to overheat and triggers negative effects.

Programming part
• Implemented a system that tracks and preserves pressure/state in the locomotive pipes over time.
• Added pipe degradation logic: every overheating event reduces pipe condition, increasing deceleration and lowering coal effectiveness.
• Implemented a repair tool (sledgehammer) mechanic:
– Pipes are repaired on impact.
– Repair only triggers if the hit has sufficient swing/impact speed.
– A spark effect is spawned at the hit point and the player camera shakes.
– The hammer can only be held with two hands; otherwise it cannot be picked up or immediately drops.
• Added an engine overheating system: when the furnace is fed too aggressively, the engine enters an overheating state and applies a camera shake effect.
• Improved train movement simulation: speed preservation and acceleration logic were refined and linked with wheel animation.
• Wheel rotation is now synchronized with actual movement speed so that wheel revolutions match the traveled distance.

Graphics part
• Added visual feedback for repair actions: sparks at the impact point on pipes.
• Added player feedback during overheating and repairs via camera shake (as a readable gameplay effect).

Plans for the near future
Game design
• Balance coal efficiency vs. pipe damage so that overheating creates meaningful risk/reward decisions.

• Add clearer player-facing indicators for pipe condition and overheating state.

• Add weapon’s recoil and physical simulation

•Weapon damage balance

Programming part

- Add UI/feedback elements to show pipe integrity, overheating level, and repair effectiveness.
- Improve tuning for repair detection (minimum hit speed) and interaction reliability in VR.
- Finalize integration of overheating, pipe condition, and train movement for consistent difficulty scaling.

Graphics part

- Improve VFX for overheating (heat shimmer, smoke, glow) and for pipe damage states (visual cracks/leaks).
- Add clearer visual markers for repairable pipe interaction points in first person/VR.

Time allocation

- Volodymyr Zaikin total 20 hours
- Aleksandr Zhidkov total 20 hours